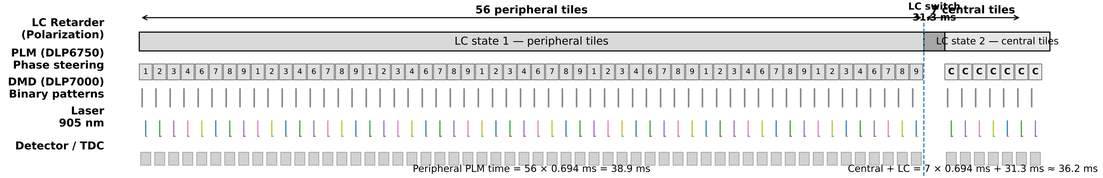
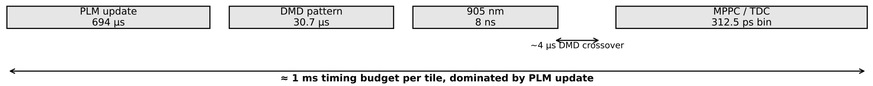
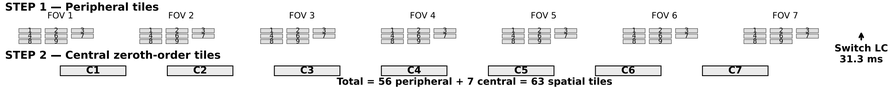
Timing and Scan Sequence of the 7 × 9 DMD–PLM Hybrid ToF LiDAR



Timing detail for one LiDAR tile



Proposed linear scan strategy: group the 56 peripheral tiles, then acquire the 7 central tiles



Parameter	Current raster	Proposed linear
PLM states	63	63
Peripheral states	56	56
Central states	7	7
LC switches	7	1
LC time	219.1 ms	31.3 ms
PLM time	43.7 ms	43.7 ms
Idealized total	~262.8 ms	~75.0 ms
Idealized scan rate	~3.8 Hz	~13.3 Hz

Fig. 1. Timing diagram and scan sequence of the 7 × 9 DMD–PLM hybrid ToF LiDAR system. The DLP7000 provides binary-pattern switching, while the DLP6750 PLM performs intra-FOV phase steering; the LCC1221-B polarization retarder is required only for the central zeroth-order tile. The proposed linear acquisition groups the 56 peripheral tiles followed by the seven central tiles, reducing the LC switching events from seven to one and substantially reducing the polarization-switching contribution to the total scan time.